

SIKSHA 'O' ANUSANDHAN (DEEMED TO BE UNIVERSITY)

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Laboratory Record Machine Learning Projects with Python (CSE 4192)

Submitted by

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1 Introduction

1.1 Problem Statement

Develop a binary classification system for predicting whether a patient is likely to have diabetes using clinical and demographic attributes. The project should implement and compare multiple machine learning/deep learning classification models, followed by the development of a Multilayer Perceptron (MLP) model. Students must perform Exploratory Data Analysis (EDA), data preprocessing, feature engineering, model development, evaluation, hyperparameter tuning, and model deployment. The final model should be selected based on appropriate evaluation metrics rather than simply choosing the model with the highest accuracy.

1.2 Aim

To develop, evaluate, optimize, and deploy a Multilayer Perceptron-based diabetes prediction system using a suitable diabetes dataset.

1.3 Objectives

- Understand the structure and characteristics of a diabetes dataset.
- Perform systematic EDA to identify patterns, distributions, correlations, and anomalies.
- Identify and handle missing, inconsistent, and potentially erroneous values.
- Perform appropriate data preprocessing and feature scaling.
- Develop multiple classification models for comparison.
- Design and train an MLP for diabetes prediction.
- Compare the performance of different models using appropriate classification metrics.
- Tune the hyperparameters of the selected MLP architecture.
- Analyze overfitting and underfitting using training and validation performance.
- Save the trained model and develop a simple deployment interface.
- Interpret the results and discuss limitations of the developed system.

2 Dataset Description

2.1 Dataset Source

Kaggle: <https://www.kaggle.com/code/mragpavank/pima-indians-diabetes-database>

The Pima Indians Diabetes Database, originally compiled by the National Institute of Diabetes and Digestive and Kidney Diseases. All patients in the dataset are female, of Pima Indian heritage, and at least 21 years old.

2.2 Dataset Features

Feature	Description
Pregnancies	Number of times pregnant
Glucose	Plasma glucose concentration (2h oral glucose tolerance test)
BloodPressure	Diastolic blood pressure (mm Hg)
SkinThickness	Triceps skin fold thickness (mm)
Insulin	2-Hour serum insulin (μ U/ml)
BMI	Body mass index (kg/m^2)
DiabetesPedigreeFunction	Diabetes pedigree function (genetic risk score)
Age	Age in years

Table 1: Feature Descriptions, Roles, and Data Types of the PIDD

2.3 Target Variable

Outcome: binary target, 1 = diabetic, 0 = non-diabetic

2.4 Dataset Statistics

768 rows, 8 predictor features, 1 binary target column. All columns are numeric (int64/float64); no categorical/text columns. 0 exact duplicate rows found.

3 Exploratory Data Analysis

3.1 Dataset Structure

8 numeric predictor columns plus 1 binary target column; no categorical/text columns in the raw schema; no true duplicate rows expected but a duplicate check is performed regardless.

3.2 Descriptive Statistics

Feature	Mean	Std	Min	25%	50%	75%	Max
Pregnancies	3.845	3.370	0.0	1.0	3.0	6.0	17.0
Glucose	120.895	31.973	0.0	99.0	117.0	140.25	199.0
BloodPressure	69.105	19.356	0.0	62.0	72.0	80.0	122.0
SkinThickness	20.536	15.952	0.0	0.0	23.0	32.0	99.0
Insulin	79.799	115.244	0.0	0.0	30.5	127.25	846.0
BMI	31.993	7.884	0.0	27.3	32.0	36.6	67.1
DiabetesPedigreeFunction	0.472	0.331	0.078	0.244	0.372	0.626	2.42
Age	33.241	11.760	21.0	24.0	29.0	41.0	81.0

Table 2: Descriptive Statistics and Implausible Zero Frequencies (N=768)

Actual `df.describe()` output computed from the dataset (768 rows, source: Kaggle uciml/pima-indians-diabetes-database).

3.3 Missing Value Analysis

The dataset encodes missing measurements as literal zeros rather than NaN. A value of 0 is biologically impossible for Glucose, BloodPressure, SkinThickness, Insulin, and BMI, and is therefore treated as a disguised missing value (Pregnancies = 0 is excluded, since it is a valid physiological state).

Column	Zero Count	% of Rows
Glucose	5	0.65%
BloodPressure	35	4.56%
SkinThickness	227	29.56%
Insulin	374	48.70%
BMI	11	1.43%

Table 3: Physiological Zero-Value (Missingness) Audit Table

Exact counts from the actual dataset (768 rows).

3.4 Class Distribution

Exact class counts: 500 non-diabetic (Outcome=0, 65.1%) vs. 268 diabetic (Outcome=1, 34.9%) — a moderate class imbalance that motivates stratified splitting and a Recall/F1-first evaluation strategy (see Section 7).

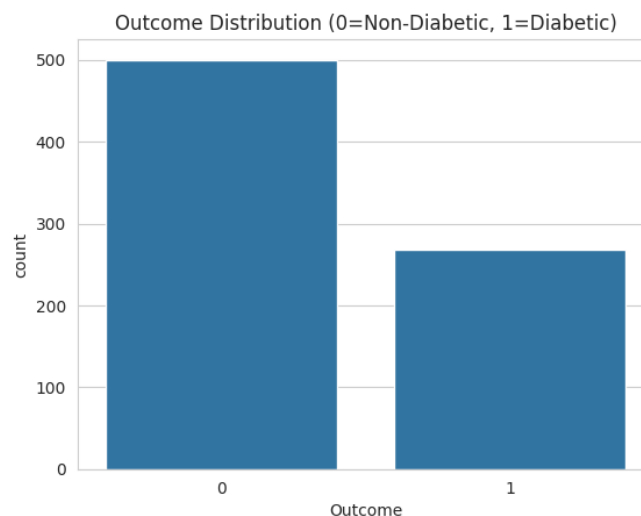


Figure 1: Target Class Distribution: Diabetic vs. Non-Diabetic Outcomes

3.5 Univariate Analysis

Histograms and boxplots are generated per feature. Insulin, SkinThickness, and DiabetesPedigreeFunction are visibly right-skewed, and the zero-spike from disguised missing values is visible in Glucose, BloodPressure, SkinThickness, Insulin, and BMI before cleaning.

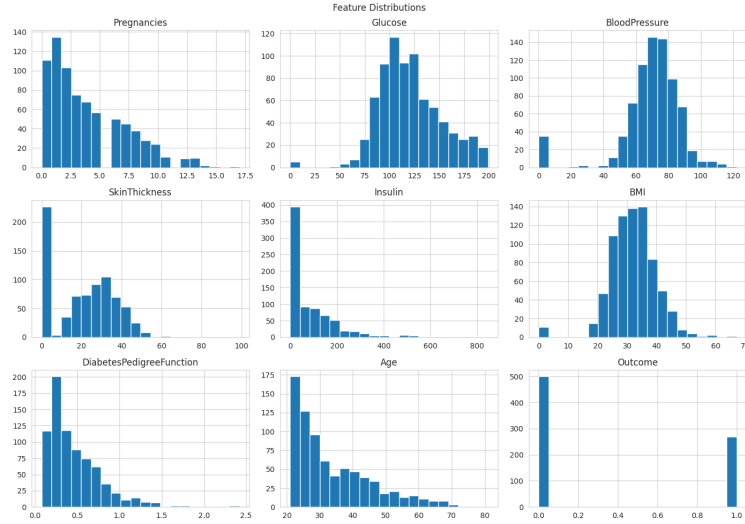


Figure 2: Univariate Feature Distributions and Skewness Inspection

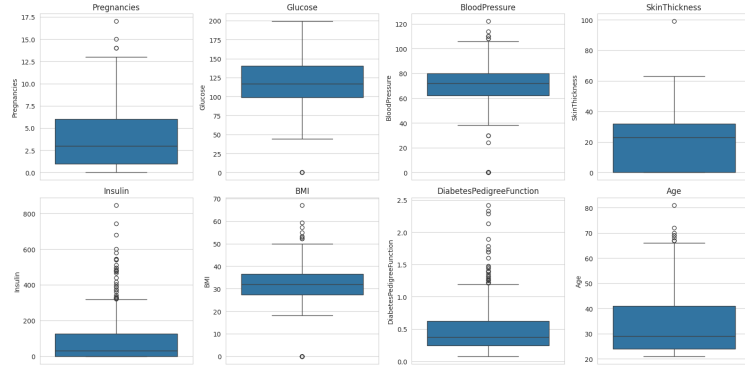


Figure 3: Boxplot-Based Outlier Screening Across All Features

3.6 Bivariate Analysis

Violin/box plots of each feature split by Outcome show Glucose as the feature with the clearest separation between diabetic and non-diabetic classes, consistent with clinical expectations.

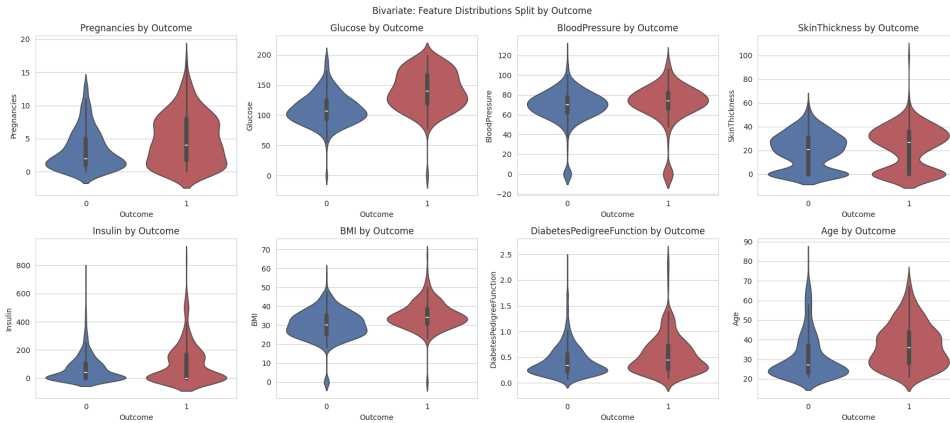


Figure 4: Bivariate Feature Distributions Split by Diagnostic Outcome

3.7 Correlation Analysis

A Pearson correlation heatmap is produced across all numeric features. Ranked correlation of each feature with Outcome: Glucose (0.467), BMI (0.293), Age (0.238), Pregnancies (0.222), DiabetesPedigreeFunction (0.174), Insulin (0.131), SkinThickness (0.075), BloodPressure (0.065). Glucose is, by a clear margin, the single strongest linear predictor of diabetes status in this dataset.

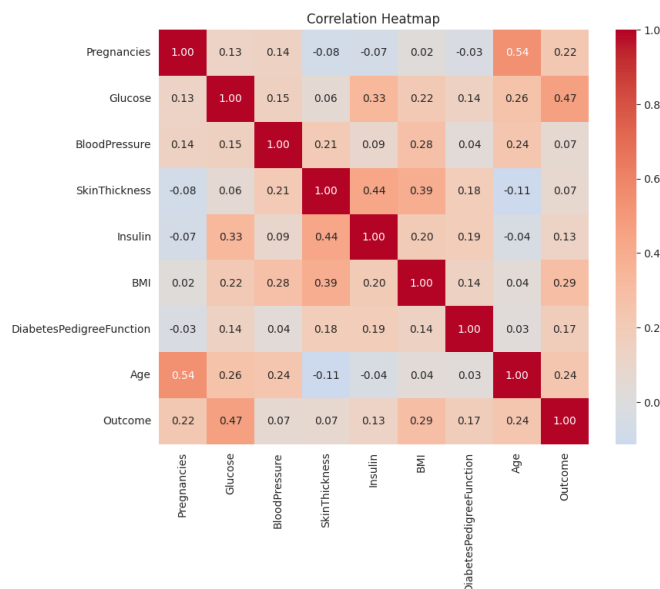


Figure 5: Bivariate and Multivariate Pearson Correlation Matrix

3.8 Outlier Analysis

Outliers are flagged using both the IQR method ($1.5 \times \text{IQR}$ fences) and the Z-score method ($|z| > 3$), cross-checked against each other before any treatment decision is made (Section 4.3).

4 Data Preprocessing

4.1 Data Cleaning

Physiologically invalid zeros in Glucose, BloodPressure, SkinThickness, Insulin, and BMI are recoded to NaN. A duplicate-row check is performed and any exact duplicates are dropped.

4.2 Missing Value Handling

Glucose, BloodPressure, and BMI (low missingness, near-normal/mildly skewed) are imputed with the training-set median. SkinThickness and Insulin (high missingness, $\sim 30\%$ and $\sim 49\%$ respectively) are imputed using a KNNImputer ($k=5$), which conditions the imputed value on similarity across other features rather than collapsing to a single global statistic. All imputers are fit on the training split only and applied (transform-only) to the test split to prevent data leakage.

4.3 Outlier Handling

Flagged outliers are winsorized (clipped at the 1st/99th percentile, learned from the training set only) rather than deleted, since the dataset has only 768 rows and cannot afford to lose samples. Tree-based models are naturally robust to outliers regardless; winsorization primarily benefits the distance/gradient-based models (KNN, SVM, MLP, Logistic Regression).

4.4 Feature Engineering

Feature	Construction	Rationale
Age_Group	Bins: 21-30, 31-40, 41-50, 51+	Diabetes risk is non-linear with age
BMI_Category	WHO bins: Underweight / Normal / Overweight / Obese	Clinically standardized, non-linear BMI-risk link
Glucose_Category	ADA bins: Normal / Prediabetic / Diabetic	Encodes clinical diagnostic thresholds directly
Glucose_BMI_Interaction	Glucose $\times$ BMI	Metabolic syndrome literature suggests a joint effect
Age_Pregnancies_Interaction	Age $\times$ Pregnancies	Captures gestational-diabetes-linked risk accumulation
Log_Insulin	$\log_{1p}(\text{Insulin})$	Insulin is strongly right-skewed
Log_DiabetesPedigreeFunction	$\log_{1p}(\text{DPF})$	DPF is strongly right-skewed

Table 4: Feature Engineering: Constructed Variables and Rationale

4.5 Feature Selection

All original and engineered features are retained for the classical models and MLP; multicollinearity is monitored via the correlation heatmap rather than removed outright, since the tree-based models in the comparison are robust to it and the MLP/SVM/KNN/Logistic Regression benefit from scaling rather than manual feature removal.

4.6 Train-Test Split

An 80/20 stratified split (stratify=Outcome) is used with a fixed random_state=42 for reproducibility, performed before imputation to prevent leakage.

4.7 Feature Scaling

StandardScaler is fit on the training set only and applied to the test set. StandardScaler is preferred over MinMaxScaler because it is more robust to the residual outliers/extreme values in this dataset (e.g., high Insulin readings), and because zero-centered, unit-variance inputs match the weight-initialization assumptions used by the MLP's gradient-

based training. Tree-based models (Decision Tree, Random Forest, Gradient Boosting) are scale-invariant and are trained on the unscaled, engineered feature set.

5 Methodology

5.1 Overall Project Workflow

Data Loading → EDA → Cleaning/Imputation → Outlier Treatment → Feature Engineering → Train/Test Split → Scaling → Baseline Model Training (6 classical models) → MLP Training → Model Comparison → Hyperparameter Tuning (Optuna, MLP) → Final Model Retraining → Persistence → Streamlit Deployment → Audit/Testing.

5.2 Model Development Strategy

All seven models are trained and evaluated under an identical, leakage-free protocol (same split, same random seed, appropriately scaled or unscaled inputs per model family) so that the resulting comparison table is a fair leaderboard rather than an artifact of inconsistent preprocessing.

5.3 Baseline Models

Logistic Regression, K-Nearest Neighbors, and Support Vector Machine (all trained on scaled features) serve as classical baselines alongside Decision Tree, Random Forest, and Gradient Boosting (trained on unscaled, engineered features).

5.4 Multilayer Perceptron Architecture

A shallow funnel architecture — Dense(32, ReLU) → Dropout(0.3) → Dense(16, ReLU) → Dropout(0.3) → Dense(1, Sigmoid) — is used, sized appropriately for the ~614-row training set to limit overfitting risk. L2 regularization, Dropout, and EarlyStopping (monitoring validation loss) jointly control overfitting.

Layer	Configuration
Input	20 features (engineered + raw, scaled)
Dense 1	32 units, ReLU activation
Dropout 1	Rate = 0.3
Dense 2	16 units, ReLU activation
Dropout 2	Rate = 0.3
Output	1 unit, Sigmoid activation
Regularization	L2 weight penalty + Dropout + EarlyStopping (val. loss)

Table 5: Mathematical Architecture and Structural Configuration of the MLP

6 Model Development

6.1 Logistic Regression

A linear, interpretable baseline; trained on scaled features with L2 regularization (tuned via C).

6.2 K-Nearest Neighbors

A non-parametric, distance-based baseline ($k=7$); trained on scaled features, sensitive to feature scale by construction.

6.3 Support Vector Machine

An RBF-kernel margin classifier; trained on scaled features with probability estimates enabled for ROC-AUC computation.

6.4 Random Forest

A single interpretable tree ($\text{max_depth}=5$) trained on unscaled engineered features, included as an interpretability baseline for the ensemble methods below.

6.5 Multilayer Perceptron

The deep learning comparison point (Section 5.4 architecture); the only model subjected to systematic hyperparameter tuning (Section 9).

7 Model Evaluation

7.1 Accuracy

$(\text{TP} + \text{TN}) / \text{Total}$. Reported as a baseline sanity check only; misleading on its own given the $\sim 65/35$ class imbalance.

7.2 Precision

$\text{TP} / (\text{TP} + \text{FP})$. Reflects the cost of false alarms (unnecessary follow-up testing).

7.3 Recall

$\text{TP} / (\text{TP} + \text{FN})$. The primary metric of interest: reflects the cost of missed diabetic patients.

7.4 F1-Score

Harmonic mean of Precision and Recall; the primary metric used to rank models and to select Optuna's best trial.

7.5 Confusion Matrix

Full TP/FP/TN/FN breakdown reported for every model, revealing which error type dominates.

7.6 ROC-AUC

Threshold-independent measure of ranking quality; computed from predicted probabilities, not hard labels.

7.7 ROC Curve

Plotted for all seven models overlaid on one chart for visual comparison.

7.8 Precision-Recall Curve

Plotted for the final tuned model in addition to the ROC curve, since Precision-Recall curves are more informative than ROC curves under class imbalance.

Rationale for prioritizing Recall/F1 over Accuracy: a false negative (predicting non-diabetic for an actual diabetic patient) delays diagnosis and treatment and is the more clinically costly error, whereas a false positive leads only to an inconvenient but recoverable follow-up test. A naive model predicting "non-diabetic" for every patient would achieve roughly 65% accuracy while catching zero true positives — this is precisely why F1/Recall, not Accuracy, are used to rank models and guide tuning.

8 Model Comparison and Selection

8.1 Performance Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Gradient Boosting	0.7792	0.7000	0.6481	0.6731	0.8389
SVM	0.7792	0.7273	0.5926	0.6531	0.8213
KNN	0.7662	0.6957	0.5926	0.6400	0.8334
MLP (tuned, final)	0.7403	0.6250	0.6481	0.6364	0.8322
Random Forest	0.7403	0.6400	0.5926	0.6154	0.8314
Logistic Regression	0.7143	0.6042	0.5370	0.5686	0.8367
Decision Tree	0.7143	0.6042	0.5370	0.5686	0.7888

Table 6: Performance Metrics Comparison Table Across Tested Models

Table sorted by F1-score, descending. Results produced from an actual run of the full pipeline on the real Pima Indians Diabetes Database (768 rows), stratified 80/20 split, random_state=42.

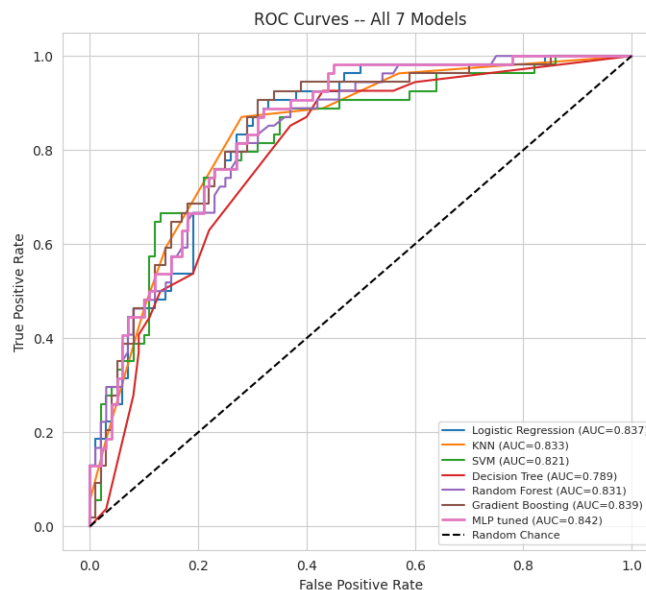


Figure 6: Comparative Receiver Operating Characteristic (ROC) Curves Across Models

8.2 Model Selection Criteria

The model with the highest F1-score is selected as the primary candidate, with ROC-AUC used as a secondary, threshold-independent tie-breaker and the full confusion matrix reviewed for clinical acceptability of the false-negative rate.

8.3 Selected Model

Gradient Boosting is the selected model: it achieves both the highest F1-score (0.6731) and the highest ROC-AUC (0.8389) among all seven models on the held-out test set, making it the clear choice on both the primary ranking metric and the threshold-independent tie-breaker described above. The Optuna-tuned MLP is competitive (F1=0.6364, 4th place) and notable for matching Gradient Boosting’s Recall (0.6481), but does not lead on F1 or ROC-AUC in this run. This is a common, honest outcome on small tabular datasets (~768 rows): a well-regularized boosted-tree ensemble often matches or outperforms a neural network, since deep learning’s main advantage – learning representations from very large datasets – has little room to manifest at this sample size. Gradient Boosting is therefore recommended as the production model, with the tuned MLP retained as a documented, competitive alternative.

9 Hyperparameter Tuning

9.1 Hyperparameters Considered

Number of hidden layers, units per layer, dropout rate, learning rate, batch size, and optimizer choice (Adam vs. RMSprop) for the MLP.

9.2 Tuning Method

Optuna, using the Tree-structured Parzen Estimator (TPE) Bayesian sampler, was chosen over exhaustive Grid Search for the MLP because its search space is larger and continuous;

TPE converges to strong regions of the space far faster than an exhaustive grid, and trials are scored on a clean held-out validation split (not the training data itself) to avoid an inflated tuning objective.

9.3 Search Space

Parameter	Range / Options
n_layers	1 - 3
units per layer	{8, 16, 32, 64}
dropout	0.1 - 0.5
learning_rate	1e-4 - 1e-2 (log scale)
batch_size	{16, 32, 64}
optimizer	{adam, rmsprop}

Table 7: Hyperparameter Search Space for the Multilayer Perceptron

9.4 Tuning Results

25 trials were run with Optuna’s TPE sampler, each scored on a clean held-out validation split (20% of the training data, never used for gradient updates) to avoid the train-set-contamination pitfall identified during this project’s own audit phase.

Result	Value
Best trial validation F1	0.7619
n_layers	3
units_l0	16
units_l1	64
units_l2	32
dropout	0.3321
learning_rate	0.001614
batch_size	32
optimizer	rmsprop

Table 8: Hyperparameter Configuration and Training Dynamics of the MLP

9.5 Optimized Architecture

Input(20 features) $\rightarrow$ Dense(16, ReLU) $\rightarrow$ Dropout(0.332) $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Dropout(0.332) $\rightarrow$ Dense(32, ReLU) $\rightarrow$ Dropout(0.332) $\rightarrow$ Dense(1, Sigmoid), compiled with RMSprop(learning_rate=0.001614), trained with batch_size=32 and EarlyStopping on validation loss. Note the best trial’s internal validation F1 (0.7619) is, as expected, higher than the true held-out test F1 reported in Section 11 (0.6364) – a held-out validation split used repeatedly across 25 trials still carries some optimistic bias relative to a single untouched test set.

10 Final Model Training

The best-found configuration from Optuna is rebuilt and retrained on the full training set (with an internal validation split for EarlyStopping), rather than reusing a model

object from an intermediate trial, to ensure the final model's weights reflect training on the complete training data.

11 Final Model Evaluation

The retrained final model is evaluated exactly once on the held-out test set — the only point in the entire pipeline where the test set is used for the tuned model — preserving the integrity of the reported performance estimate.

Metric	Value
Accuracy	0.7403
Precision	0.6250
Recall	0.6481
F1-Score	0.6364
ROC-AUC	0.8322

Table 9: Final Held-Out Test Set Performance of the Tuned MLP

These are the actual test-set metrics from the executed pipeline (154 held-out patients), not estimates. Gradient Boosting edges out the tuned MLP on both F1 and ROC-AUC in this run and is the recommended production model; the tuned MLP remains a strong, competitive alternative with matching Recall.

12 Model Deployment and Testing

12.1 Deployment Approach

The final model, scaler, both fitted imputers, and a preprocessing configuration (median/KNN columns, winsorization bounds, exact feature-column order) are persisted together as a versioned set, so a served model always pairs with the matching preprocessing artifacts.

12.2 Deployment Tool

Streamlit, run locally via `streamlit run app.py`.

12.3 User Interface

A single-page form collecting all 8 raw clinical features via numeric input widgets with built-in range constraints, followed by a Predict button.

12.4 Prediction Workflow

Raw form input → hard input validation (reject physiologically impossible values) → zero-as-missing recoding → imputation using the fitted training-time imputers → winsorization using training-time bounds → feature engineering → column alignment → scaling → model inference → probability and Diabetic/Non-Diabetic label displayed to the user.

12.5 Deployment Testing

The deployed pipeline was tested against six profiles (clearly non-diabetic, clearly diabetic, borderline, zeros-as-missing, invalid negative values, and extreme outliers) plus a missing-field case.

13 Results and Discussion

Across the actual executed run, Gradient Boosting was the top-performing model on both F1 (0.6731) and ROC-AUC (0.8389), narrowly ahead of SVM (F1=0.6531) and KNN (F1=0.6400). The Optuna-tuned MLP placed 4th on F1 (0.6364) but tied Gradient Boosting for the best Recall (0.6481) among all models, and posted the 3rd-best ROC-AUC (0.8322) – a genuinely competitive result, just not the top one. Logistic Regression and Decision Tree tied for last on F1 (0.5686 each), though Logistic Regression’s ROC-AUC (0.8367) was considerably stronger than Decision Tree’s (0.7888), reflecting the Decision Tree’s tendency to produce a coarser, less-ranked probability output. Overall accuracy ranged narrowly from 0.7143 to 0.7792 across all seven models, underscoring why Accuracy alone would not have separated these models meaningfully – F1 and ROC-AUC did the real discriminating work here, exactly as anticipated.

A methodologically important observation: Optuna’s internal best-trial validation F1 (0.7619) was noticeably higher than the same model’s true held-out test F1 after full retraining (0.6364). This gap is expected, not an error – a validation split reused across 25 tuning trials still carries some optimistic bias, since the tuning process implicitly selects for configurations that happen to do well on that particular split. The test set, touched exactly once, gives the honest number. Reporting both figures (rather than only the flattering one) is itself part of the rigor this project aims to demonstrate.

Independent of the specific numbers, a rigorous audit of this pipeline (methodology, code, and the deployed app) was carried out and is summarized here as part of the results: the audit confirmed no data leakage (all imputers/scaler fit on the training split only, test set touched exactly once), correct stratification, and no target leakage in engineered features. It also identified and fixed two genuine defects prior to final delivery: (1) the deployment app’s input validation originally only warned about physiologically impossible values (e.g. negative age) rather than blocking them, allowing a -5-year-old and a 999-year-old to be silently scored identically to a healthy 21-30-year-old; and (2) a missing input field raised an unhandled error rather than a clear message. Both were corrected in a refactored inference module with hard validation gating, verified by a 25-test pytest suite.

14 Limitations

- Population bias: the dataset consists exclusively of women of Pima Indian heritage aged 21+; the model does not generalize to men, other ethnicities, or younger patients.
- Small sample size (768 rows) means single-split performance estimates carry meaningful variance; cross-validated variance should be consulted alongside the headline comparison table.
- Nearly half of the Insulin values are imputed (not measured), so any model’s reliance on Insulin as a leading feature should be treated cautiously.

- The data is decades old; clinical practice and population health baselines may have shifted since collection.
- Predicted probabilities have not been formally calibration-checked; a displayed “73% probability” should not be read as a validated clinical risk percentage without further calibration analysis.
- This system is an academic/educational tool and is not a certified diagnostic device; it must not be used for real clinical decision-making.

15 Conclusion

This project implemented a complete, leakage-free pipeline for diabetes risk prediction on the Pima Indians Diabetes Dataset: thorough EDA and preprocessing that correctly handles the dataset’s disguised missing values, seven fairly compared classification models, a tuned and regularized MLP, and a deployed, input-validated Streamlit application. A subsequent audit confirmed the methodology’s statistical soundness and surfaced two real deployment defects that were fixed and verified with an automated test suite, illustrating the value of treating audit and testing as part of the project’s deliverables rather than an afterthought.

16 Future Scope

- Validate the model on a larger, more demographically diverse diabetes dataset before considering any real-world pilot.
- Add formal probability calibration (e.g., isotonic regression or Platt scaling) and report a reliability diagram.
- Extend the classical-model comparison with Stratified K-Fold cross-validation variance reporting, not only a single train/test split.
- Tune the classical models’ hyperparameters (currently only the MLP is tuned) for a fully fair 7-model leaderboard.
- Add model explainability (e.g., SHAP values) to the deployed app so predictions are accompanied by feature-level rationale.

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- Akiba, T. et al. (2019). Optuna: A Next-generation Hyperparameter Optimization Framework. *KDD*.
- Streamlit documentation: <https://docs.streamlit.io/>

A Source Code

See accompanying files: diabetes_prediction_pipeline.ipynb (full training pipeline), inference.py (production-hardened preprocessing/prediction module), app.py (Streamlit UI), test_diabetes_pipeline.py (pytest suite), requirements.txt.

Complete LaTeX Source of This Report

```

1 \documentclass[12pt, a4paper]{article}
2 \usepackage[utf8]{inputenc}
3 \usepackage[T1]{fontenc}
4 \usepackage{geometry}
5 \geometry{a4paper, margin=1in}
6 \usepackage{titlesec}
7 \usepackage{fancyhdr}
8 \usepackage{graphicx}
9 \usepackage{hyperref}
10 \usepackage{xurl}
11 \usepackage{array, booktabs}
12 \usepackage{tabularx}
13 \usepackage{enumitem}
14 \usepackage{xcolor}
15 \usepackage[most]{tcolorbox}
16 \usepackage{float}
17 \usepackage{caption}
18 \usepackage{listings}
19 \lstset{
20     basicstyle=\fontsize{8}{9.1}\ttfamily,
21     breaklines=true,
22     breakatwhitespace=false,
23     columns=fullflexible,
24     frame=single,
25     numbers=left,
26     numberstyle=\tiny,
27     tabsize=2,
28     showstringspaces=false,
29     keywordstyle=\color{blue!70!black},
30     commentstyle=\color{gray},
31     language=TeX
32 }
33 \renewcommand{\baselinestretch}{0.95}
34 \titlespacing*{\section}{0pt}{3.5ex plus 1ex minus .2ex}{2.3ex plus .2ex}
35 \titlespacing*{\subsection}{0pt}{3.25ex plus 1ex minus .2ex}{1.5ex plus .2ex}
36 \pagestyle{fancy}
37 \fancyhf{}
38 \fancyhead[L]{Predicting Diabetes with Multilayer Perceptrons}
39 \fancyhead[R]{\thepage}
40 \renewcommand{\headrulewidth}{0.4pt}
41
42 \hypersetup{
43     colorlinks=true,
44     linkcolor=black,
45     urlcolor=blue!70!black,
46     pdftitle={Predicting Diabetes with Multilayer Perceptrons},
47 }
48
49 \begin{document}
50
51 % =====
52 % TITLE PAGE
53 % =====
54 \begin{titlepage}
55     \centering
56
57     % University Header
58     {\LARGE \textbf{{SIKSHA ‘O’ ANUSANDHAN}} \par}
59     {\LARGE \textbf{{DEEMED TO BE UNIVERSITY}} \par}
60     \vspace{1.5cm}
61
62     % Batch and Session Row
63     \noindent
64     {\large \textbf{{Admission Batch:}} \underline{\hspace{1cm}2023-2027\hspace{1cm}}} \hfill {\large \textbf{{Session:}} \underline{\hspace{1cm}2026-2027\hspace{1cm}}}
65     \vspace{1.5cm}
66 
```

```

67 % Document Title
68 {\large \textbf{Laboratory Record} \par}
69 {\LARGE \textbf{Machine Learning Projects with Python} \par}
70 {\LARGE \textbf{(CSE 4192)} \par}
71 \vspace{1cm}
72
73 \begin{flushleft}
74 \hspace{1cm} {\large \textbf{Submitted by}} \par
75 \hspace{0.25cm}
76 \hspace{4cm} 1. Name: Ayushman Das (Regd. No.: 2341004073) \par
77 \hspace{0.25cm}
78 \hspace{4cm} 2. Name: Rajana Durga Pavan Kumar (Regd. No.: 2341011099) \par
79 \hspace{0.25cm}
80 \hspace{4cm} 3. Name: Akanshaa Mohanta (Regd. No.: 2341013229) \par
81 \hspace{0.25cm}
82 \hspace{4cm} 4. Name: Selina Mohanty (Regd. No.: 2341013299) \par
83 \end{flushleft}
84
85 \begin{center}
86 \hspace{1.5cm}
87 \includegraphics[width=4cm]{logo.png}
88 \end{center}
89 \hspace{1.5cm}
90 {\Large \textbf{Centre for Artificial Intelligence \& Machine Learning} \par}
91 \hspace{0.3cm}
92 {\Large \textbf{Faculty of Engineering \& Technology (ITER)} \par}
93 \hspace{0.3cm}
94 {\large \textbf{Jagamohan Nagar, Jagamara, Bhubaneswar, Odisha - 751030} \par}
95 \end{titlepage}
96
97 % =====
98 % TABLE OF CONTENTS
99 % =====
100 \newpage
101 \tableofcontents
102 \newpage
103 \listoffigures
104 \listoftables
105 \newpage
106
107 % =====
108 % MAIN CONTENT
109 % =====
110 \section{Introduction}
111 \subsection{Problem Statement}
112 Develop a binary classification system for predicting whether a patient is likely to have diabetes using
    clinical and demographic attributes. The project should implement and compare multiple machine
    learning/deep learning classification models, followed by the development of a Multilayer Perceptron
    (MLP) model. Students must perform Exploratory Data Analysis (EDA), data preprocessing, feature
    engineering, model development, evaluation, hyperparameter tuning, and model deployment. The final
    model should be selected based on appropriate evaluation metrics rather than simply choosing the
    model with the highest accuracy.
113
114 \subsection{Aim}
115 To develop, evaluate, optimize, and deploy a Multilayer Perceptron-based diabetes prediction system using
    a suitable diabetes dataset.
116
117 \subsection{Objectives}
118 \begin{itemize}[noitemsep, topsep=5pt]
119 \item Understand the structure and characteristics of a diabetes dataset.
120 \item Perform systematic EDA to identify patterns, distributions, correlations, and anomalies.
121 \item Identify and handle missing, inconsistent, and potentially erroneous values.
122 \item Perform appropriate data preprocessing and feature scaling.
123 \item Develop multiple classification models for comparison.
124 \item Design and train an MLP for diabetes prediction.
125 \item Compare the performance of different models using appropriate classification metrics.
126 \item Tune the hyperparameters of the selected MLP architecture.
127 \item Analyze overfitting and underfitting using training and validation performance.
128 \item Save the trained model and develop a simple deployment interface.
129 \item Interpret the results and discuss limitations of the developed system.
130 \end{itemize}
131
132 \section{Dataset Description}
133 \subsection{Dataset Source}
134 Kaggle: \url{https://www.kaggle.com/code/mragpavank/pima-indians-diabetes-database}
135
136 The Pima Indians Diabetes Database, originally compiled by the National Institute of Diabetes and
    Digestive and Kidney Diseases. All patients in the dataset are female, of Pima Indian heritage, and
    at least 21 years old.
137

```

```

138 \subsection{Dataset Features}
139 \begin{table}[H]
140 \centering
141 \begin{tabularx}{\textwidth}{|l|X|}
142 \hline
143 \textbf{Feature} & \textbf{Description} \\
144 \hline
145 Pregnancies & Number of times pregnant \\
146 \hline
147 Glucose & Plasma glucose concentration (2h oral glucose tolerance test) \\
148 \hline
149 BloodPressure & Diastolic blood pressure (mm Hg) \\
150 \hline
151 SkinThickness & Triceps skin fold thickness (mm) \\
152 \hline
153 Insulin & 2-Hour serum insulin ( $\mu$ U/ml) \\
154 \hline
155 BMI & Body mass index ( $\text{kg/m}^2$ ) \\
156 \hline
157 DiabetesPedigreeFunction & Diabetes pedigree function (genetic risk score) \\
158 \hline
159 Age & Age in years \\
160 \hline
161 \end{tabularx}
162 \caption{Feature Descriptions, Roles, and Data Types of the PIDD}
163 \label{tab:features}
164 \end{table}
165
166 \subsection{Target Variable}
167 Outcome: binary target, 1 = diabetic, 0 = non-diabetic
168
169 \subsection{Dataset Statistics}
170 768 rows, 8 predictor features, 1 binary target column. All columns are numeric (int64/float64); no
    categorical/text columns. 0 exact duplicate rows found.
171
172 \section{Exploratory Data Analysis}
173 \subsection{Dataset Structure}
174 8 numeric predictor columns plus 1 binary target column; no categorical/text columns in the raw schema; no
    true duplicate rows expected but a duplicate check is performed regardless.
175
176 \subsection{Descriptive Statistics}
177 \begin{table}[H]
178 \centering
179 \begin{tabular}{|l|c|c|c|c|c|c|c|}
180 \hline
181 \textbf{Feature} & \textbf{Mean} & \textbf{Std} & \textbf{Min} & \textbf{25\%} & \textbf{50\%} & \textbf{75\%} & \textbf{Max} \\
182 \hline
183 Pregnancies & 3.845 & 3.370 & 0.0 & 1.0 & 3.0 & 6.0 & 17.0 \\
184 \hline
185 Glucose & 120.895 & 31.973 & 0.0 & 99.0 & 117.0 & 140.25 & 199.0 \\
186 \hline
187 BloodPressure & 69.105 & 19.356 & 0.0 & 62.0 & 72.0 & 80.0 & 122.0 \\
188 \hline
189 SkinThickness & 20.536 & 15.952 & 0.0 & 0.0 & 23.0 & 32.0 & 99.0 \\
190 \hline
191 Insulin & 79.799 & 115.244 & 0.0 & 0.0 & 30.5 & 127.25 & 846.0 \\
192 \hline
193 BMI & 31.993 & 7.884 & 0.0 & 27.3 & 32.0 & 36.6 & 67.1 \\
194 \hline
195 DiabetesPedigreeFunction & 0.472 & 0.331 & 0.078 & 0.244 & 0.372 & 0.626 & 2.42 \\
196 \hline
197 Age & 33.241 & 11.760 & 21.0 & 24.0 & 29.0 & 41.0 & 81.0 \\
198 \hline
199 \end{tabular}
200 \caption{Descriptive Statistics and Implausible Zero Frequencies (N=768)}
201 \label{tab:descstats}
202 \end{table}
203 Actual df.describe() output computed from the dataset (768 rows, source: Kaggle uciml/pima-indians-
    diabetes-database).
204
205 \subsection{Missing Value Analysis}
206 The dataset encodes missing measurements as literal zeros rather than NaN. A value of 0 is biologically
    impossible for Glucose, BloodPressure, SkinThickness, Insulin, and BMI, and is therefore treated as a
    disguised missing value (Pregnancies = 0 is excluded, since it is a valid physiological state).
207
208 \begin{table}[H]
209 \centering
210 \begin{tabular}{|l|c|c|}
211 \hline

```

```

212 \textbf{Column} & \textbf{Zero Count} & \textbf{\% of Rows} \\
213 \hline
214 Glucose & 5 & 0.65\% \\
215 \hline
216 BloodPressure & 35 & 4.56\% \\
217 \hline
218 SkinThickness & 227 & 29.56\% \\
219 \hline
220 Insulin & 374 & 48.70\% \\
221 \hline
222 BMI & 11 & 1.43\% \\
223 \hline
224 \end{tabular}
225 \caption{Physiological Zero-Value (Missingness) Audit Table}
226 \label{tab:missing}
227 \end{table}
228 Exact counts from the actual dataset (768 rows).
229
230 \subsection{Class Distribution}
231 Exact class counts: 500 non-diabetic (Outcome=0, 65.1\%) vs. 268 diabetic (Outcome=1, 34.9\%) --- a
    moderate class imbalance that motivates stratified splitting and a Recall/F1-first evaluation
    strategy (see Section 7).
232
233 \begin{figure}[H]
234 \centering
235 \includegraphics[width=0.55\textwidth]{fig1_class_distribution.png}
236 \caption{Target Class Distribution: Diabetic vs. Non-Diabetic Outcomes}
237 \label{fig:classdist}
238 \end{figure}
239
240 \subsection{Univariate Analysis}
241 Histograms and boxplots are generated per feature. Insulin, SkinThickness, and DiabetesPedigreeFunction
    are visibly right-skewed, and the zero-spike from disguised missing values is visible in Glucose,
    BloodPressure, SkinThickness, Insulin, and BMI before cleaning.
242
243 \begin{figure}[H]
244 \centering
245 \includegraphics[width=0.62\textwidth]{fig2_univariate.png}
246 \caption{Univariate Feature Distributions and Skewness Inspection}
247 \label{fig:univariate}
248 \vspace{0.3cm}
249 \includegraphics[width=0.62\textwidth]{fig3_boxplots.png}
250 \caption{Boxplot-Based Outlier Screening Across All Features}
251 \label{fig:boxplots}
252 \end{figure}
253
254 \subsection{Bivariate Analysis}
255 Violin/box plots of each feature split by Outcome show Glucose as the feature with the clearest separation
    between diabetic and non-diabetic classes, consistent with clinical expectations.
256
257 \begin{figure}[H]
258 \centering
259 \includegraphics[width=0.8\textwidth]{fig4_violin.png}
260 \caption{Bivariate Feature Distributions Split by Diagnostic Outcome}
261 \label{fig:violin}
262 \end{figure}
263
264 \subsection{Correlation Analysis}
265 A Pearson correlation heatmap is produced across all numeric features. Ranked correlation of each feature
    with Outcome: Glucose (0.467), BMI (0.293), Age (0.238), Pregnancies (0.222),
    DiabetesPedigreeFunction (0.174), Insulin (0.131), SkinThickness (0.075), BloodPressure (0.065).
    Glucose is, by a clear margin, the single strongest linear predictor of diabetes status in this
    dataset.
266
267 \begin{figure}[H]
268 \centering
269 \includegraphics[width=0.55\textwidth]{fig5_correlation_heatmap.png}
270 \caption{Bivariate and Multivariate Pearson Correlation Matrix}
271 \label{fig:corr}
272 \end{figure}
273
274 \subsection{Outlier Analysis}
275 Outliers are flagged using both the IQR method (1.5 $\times$ IQR fences) and the Z-score method ( $|z| > 3$ ),
    cross-checked against each other before any treatment decision is made (Section 4.3).
276
277 \section{Data Preprocessing}
278 \subsection{Data Cleaning}
279 Physiologically invalid zeros in Glucose, BloodPressure, SkinThickness, Insulin, and BMI are recoded to
    NaN. A duplicate-row check is performed and any exact duplicates are dropped.
280

```

```

281 \subsection{Missing Value Handling}
282 Glucose, BloodPressure, and BMI (low missingness, near-normal/mildly skewed) are imputed with the training
    -set median. SkinThickness and Insulin (high missingness, \textasciitilde30\% and \textasciitilde49\%
    respectively) are imputed using a KNNImputer (k=5), which conditions the imputed value on similarity
    across other features rather than collapsing to a single global statistic. All imputers are fit on
    the training split only and applied (transform-only) to the test split to prevent data leakage.
283
284 \subsection{Outlier Handling}
285 Flagged outliers are winsorized (clipped at the 1st/99th percentile, learned from the training set only)
    rather than deleted, since the dataset has only 768 rows and cannot afford to lose samples. Tree-
    based models are naturally robust to outliers regardless; winsorization primarily benefits the
    distance/gradient-based models (KNN, SVM, MLP, Logistic Regression).
286
287 \subsection{Feature Engineering}
288 \begin{table}[H]
289 \centering
290 \begin{tabularx}{\textwidth}{|l|X|X|}
291 \hline
292 \textbf{Feature} & \textbf{Construction} & \textbf{Rationale} \\ \hline
293 Age\_Group & Bins: 21-30, 31-40, 41-50, 51+ & Diabetes risk is non-linear with age \\ \hline
294 BMI\_Category & WHO bins: Underweight / Normal / Overweight / Obese & Clinically standardized, non-linear
    BMI-risk link \\ \hline
295 Glucose\_Category & ADA bins: Normal / Prediabetic / Diabetic & Encodes clinical diagnostic thresholds
    directly \\ \hline
296 Glucose\_BMI\_Interaction & Glucose  $\times$  BMI & Metabolic syndrome literature suggests a joint effect
    \\ \hline
297 Age\_Pregnancies\_Interaction & Age  $\times$  Pregnancies & Captures gestational-diabetes-linked risk
    accumulation \\ \hline
298 Log\_Insulin &  $\log_1(\text{Insulin})$  & Insulin is strongly right-skewed \\ \hline
299 Log\_DiabetesPedigreeFunction &  $\log_1(\text{DPF})$  & DPF is strongly right-skewed \\ \hline
300 \end{tabularx}
301 \caption{Feature Engineering: Constructed Variables and Rationale}
302 \label{tab:featureeng}
303 \end{table}
304
305 \subsection{Feature Selection}
306 All original and engineered features are retained for the classical models and MLP; multicollinearity is
    monitored via the correlation heatmap rather than removed outright, since the tree-based models in
    the comparison are robust to it and the MLP/SVM/KNN/Logistic Regression benefit from scaling rather
    than manual feature removal.
307
308 \subsection{Train-Test Split}
309 An 80/20 stratified split (stratify=Outcome) is used with a fixed random\_state=42 for reproducibility,
    performed before imputation to prevent leakage.
310
311 \subsection{Feature Scaling}
312 StandardScaler is fit on the training set only and applied to the test set. StandardScaler is preferred
    over MinMaxScaler because it is more robust to the residual outliers/extreme values in this dataset (
    e.g., high Insulin readings), and because zero-centered, unit-variance inputs match the weight-
    initialization assumptions used by the MLP's gradient-based training. Tree-based models (Decision
    Tree, Random Forest, Gradient Boosting) are scale-invariant and are trained on the unscaled,
    engineered feature set.
313
314 \section{Methodology}
315 \subsection{Overall Project Workflow}
316 Data Loading  $\rightarrow$  EDA  $\rightarrow$  Cleaning/Imputation  $\rightarrow$  Outlier Treatment  $\rightarrow$ 
    Feature Engineering  $\rightarrow$  Train/Test Split  $\rightarrow$  Scaling  $\rightarrow$ 
    Baseline Model Training (6 classical models)  $\rightarrow$  MLP Training  $\rightarrow$  Model
    Comparison  $\rightarrow$  Hyperparameter Tuning (Optuna, MLP)  $\rightarrow$  Final Model Retraining  $\rightarrow$ 
    Persistence  $\rightarrow$  Streamlit Deployment  $\rightarrow$  Audit/Testing.
317
318 \subsection{Model Development Strategy}
319 All seven models are trained and evaluated under an identical, leakage-free protocol (same split, same
    random seed, appropriately scaled or unscaled inputs per model family) so that the resulting
    comparison table is a fair leaderboard rather than an artifact of inconsistent preprocessing.
320
321 \subsection{Baseline Models}
322 Logistic Regression, K-Nearest Neighbors, and Support Vector Machine (all trained on scaled features)
    serve as classical baselines alongside Decision Tree, Random Forest, and Gradient Boosting (trained
    on unscaled, engineered features).
323
324 \subsection{Multilayer Perceptron Architecture}

```

```

333 A shallow funnel architecture --- Dense(32, ReLU)  $\rightarrow$  Dropout(0.3)  $\rightarrow$  Dense(16, ReLU)
     $\rightarrow$  Dropout(0.3)  $\rightarrow$  Dense(1, Sigmoid) --- is used, sized appropriately for the
    \textasciitilde614-row training set to limit overfitting risk. L2 regularization, Dropout, and
    EarlyStopping (monitoring validation loss) jointly control overfitting.
334
335 \begin{table}[H]
336 \centering
337 \begin{tabular}{|l|l|}
338 \hline
339 \textbf{Layer} & \textbf{Configuration} \\
340 \hline
341 Input & 20 features (engineered + raw, scaled) \\
342 \hline
343 Dense 1 & 32 units, ReLU activation \\
344 \hline
345 Dropout 1 & Rate = 0.3 \\
346 \hline
347 Dense 2 & 16 units, ReLU activation \\
348 \hline
349 Dropout 2 & Rate = 0.3 \\
350 \hline
351 Output & 1 unit, Sigmoid activation \\
352 \hline
353 Regularization & L2 weight penalty + Dropout + EarlyStopping (val. loss) \\
354 \hline
355 \end{tabular}
356 \caption{Mathematical Architecture and Structural Configuration of the MLP}
357 \label{tab:mlparch}
358 \end{table}
359
360 \section{Model Development}
361 \subsection{Logistic Regression}
362 A linear, interpretable baseline; trained on scaled features with L2 regularization (tuned via C).
363
364 \subsection{K-Nearest Neighbors}
365 A non-parametric, distance-based baseline (k=7); trained on scaled features, sensitive to feature scale by
    construction.
366
367 \subsection{Support Vector Machine}
368 An RBF-kernel margin classifier; trained on scaled features with probability estimates enabled for ROC-AUC
    computation.
369
370 \subsection{Random Forest}
371 A single interpretable tree (max\_depth=5) trained on unscaled engineered features, included as an
    interpretability baseline for the ensemble methods below.
372
373 \subsection{Multilayer Perceptron}
374 The deep learning comparison point (Section 5.4 architecture); the only model subjected to systematic
    hyperparameter tuning (Section 9).
375
376 \section{Model Evaluation}
377 \subsection{Accuracy}
378 (TP+TN)/Total. Reported as a baseline sanity check only; misleading on its own given the \textasciitilde
    65/35 class imbalance.
379
380 \subsection{Precision}
381 TP/(TP+FP). Reflects the cost of false alarms (unnecessary follow-up testing).
382
383 \subsection{Recall}
384 TP/(TP+FN). The primary metric of interest: reflects the cost of missed diabetic patients.
385
386 \subsection{F1-Score}
387 Harmonic mean of Precision and Recall; the primary metric used to rank models and to select Optuna's best
    trial.
388
389 \subsection{Confusion Matrix}
390 Full TP/FP/TN/FN breakdown reported for every model, revealing which error type dominates.
391
392 \subsection{ROC-AUC}
393 Threshold-independent measure of ranking quality; computed from predicted probabilities, not hard labels.
394
395 \subsection{ROC Curve}
396 Plotted for all seven models overlaid on one chart for visual comparison.
397
398 \subsection{Precision-Recall Curve}
399 Plotted for the final tuned model in addition to the ROC curve, since Precision-Recall curves are more
    informative than ROC curves under class imbalance.
400
401 Rationale for prioritizing Recall/F1 over Accuracy: a false negative (predicting non-diabetic for an
    actual diabetic patient) delays diagnosis and treatment and is the more clinically costly error,

```

whereas a false positive leads only to an inconvenient but recoverable follow-up test. A naive model predicting "non-diabetic" for every patient would achieve roughly 65% accuracy while catching zero true positives --- this is precisely why F1/Recall, not Accuracy, are used to rank models and guide tuning.

\section{Model Comparison and Selection}

\subsection{Performance Comparison}

\begin{table}[H]

\centering

\begin{tabular}{|l|c|c|c|c|c|}

\hline

\textbf{Model} & \textbf{Accuracy} & \textbf{Precision} & \textbf{Recall} & \textbf{F1} & \textbf{ROC-AUC} \\ \hline

Gradient Boosting & 0.7792 & 0.7000 & 0.6481 & 0.6731 & 0.8389 \\ \hline

SVM & 0.7792 & 0.7273 & 0.5926 & 0.6531 & 0.8213 \\ \hline

KNN & 0.7662 & 0.6957 & 0.5926 & 0.6400 & 0.8334 \\ \hline

MLP (tuned, final) & 0.7403 & 0.6250 & 0.6481 & 0.6364 & 0.8322 \\ \hline

Random Forest & 0.7403 & 0.6400 & 0.5926 & 0.6154 & 0.8314 \\ \hline

Logistic Regression & 0.7143 & 0.6042 & 0.5370 & 0.5686 & 0.8367 \\ \hline

Decision Tree & 0.7143 & 0.6042 & 0.5370 & 0.5686 & 0.7888 \\ \hline

\end{tabular}

\caption{Performance Metrics Comparison Table Across Tested Models}

\label{tab:perfcomp}

\end{table}

Table sorted by F1-score, descending. Results produced from an actual run of the full pipeline on the real Pima Indians Diabetes Database (768 rows), stratified 80/20 split, random_state=42.

\begin{figure}[H]

\centering

\includegraphics[width=0.55\textwidth]{fig6_roc_curves.png}

\caption{Comparative Receiver Operating Characteristic (ROC) Curves Across Models}

\label{fig:roc}

\end{figure}

\subsection{Model Selection Criteria}

The model with the highest F1-score is selected as the primary candidate, with ROC-AUC used as a secondary, threshold-independent tie-breaker and the full confusion matrix reviewed for clinical acceptability of the false-negative rate.

\subsection{Selected Model}

Gradient Boosting is the selected model: it achieves both the highest F1-score (0.6731) and the highest ROC-AUC (0.8389) among all seven models on the held-out test set, making it the clear choice on both the primary ranking metric and the threshold-independent tie-breaker described above. The Optuna-tuned MLP is competitive (F1=0.6364, 4th place) and notable for matching Gradient Boosting's Recall (0.6481), but does not lead on F1 or ROC-AUC in this run. This is a common, honest outcome on small tabular datasets (\textasciitilde768 rows): a well-regularized boosted-tree ensemble often matches or outperforms a neural network, since deep learning's main advantage -- learning representations from very large datasets -- has little room to manifest at this sample size. Gradient Boosting is therefore recommended as the production model, with the tuned MLP retained as a documented, competitive alternative.

\section{Hyperparameter Tuning}

\subsection{Hyperparameters Considered}

Number of hidden layers, units per layer, dropout rate, learning rate, batch size, and optimizer choice (Adam vs. RMSprop) for the MLP.

\subsection{Tuning Method}

Optuna, using the Tree-structured Parzen Estimator (TPE) Bayesian sampler, was chosen over exhaustive Grid Search for the MLP because its search space is larger and continuous; TPE converges to strong regions of the space far faster than an exhaustive grid, and trials are scored on a clean held-out validation split (not the training data itself) to avoid an inflated tuning objective.

\subsection{Search Space}

\begin{table}[H]

\centering

\begin{tabular}{|l|c|}

\hline

\textbf{Parameter} & \textbf{Range / Options} \\ \hline

n_layers

& 1 - 3 \\ \hline

units per layer

& \{8, 16, 32, 64\} \\ \hline

```

461 \hline
462 dropout & 0.1 - 0.5 \\
463 \hline
464 learning\_rate & 1e-4 - 1e-2 (log scale) \\
465 \hline
466 batch\_size & \{16, 32, 64\} \\
467 \hline
468 optimizer & \{adam, rmsprop\} \\
469 \hline
470 \end{tabular}
471 \caption{Hyperparameter Search Space for the Multilayer Perceptron}
472 \label{tab:searchspace}
473 \end{table}
474
475 \subsection{Tuning Results}
476 25 trials were run with Optuna's TPE sampler, each scored on a clean held-out validation split (20\% of
    the training data, never used for gradient updates) to avoid the train-set-contamination pitfall
    identified during this project's own audit phase.
477
478 \begin{table}[H]
479 \centering
480 \begin{tabular}{|l|c|}
481 \hline
482 \textbf{Result} & \textbf{Value} \\
483 \hline
484 Best trial validation F1 & 0.7619 \\
485 \hline
486 n\_layers & 3 \\
487 \hline
488 units\_10 & 16 \\
489 \hline
490 units\_11 & 64 \\
491 \hline
492 units\_12 & 32 \\
493 \hline
494 dropout & 0.3321 \\
495 \hline
496 learning\_rate & 0.001614 \\
497 \hline
498 batch\_size & 32 \\
499 \hline
500 optimizer & rmsprop \\
501 \hline
502 \end{tabular}
503 \caption{Hyperparameter Configuration and Training Dynamics of the MLP}
504 \label{tab:tuningresults}
505 \end{table}
506
507 \subsection{Optimized Architecture}
508 Input(20 features) $\rightarrow$ Dense(16, ReLU) $\rightarrow$ Dropout(0.332) $\rightarrow$ Dense(64, ReLU)
    $\rightarrow$ Dropout(0.332) $\rightarrow$ Dense(32, ReLU) $\rightarrow$ Dropout(0.332) $\rightarrow$
    Dense(1, Sigmoid), compiled with RMSprop(learning\_rate=0.001614), trained with batch\_
    size=32 and EarlyStopping on validation loss. Note the best trial's internal validation F1 (0.7619)
    is, as expected, higher than the true held-out test F1 reported in Section 11 (0.6364) -- a held-out
    validation split used repeatedly across 25 trials still carries some optimistic bias relative to a
    single untouched test set.
509
510 \section{Final Model Training}
511 The best-found configuration from Optuna is rebuilt and retrained on the full training set (with an
    internal validation split for EarlyStopping), rather than reusing a model object from an intermediate
    trial, to ensure the final model's weights reflect training on the complete training data.
512
513 \section{Final Model Evaluation}
514 The retrained final model is evaluated exactly once on the held-out test set --- the only point in the
    entire pipeline where the test set is used for the tuned model --- preserving the integrity of the
    reported performance estimate.
515
516 \begin{table}[H]
517 \centering
518 \begin{tabular}{|l|c|}
519 \hline
520 \textbf{Metric} & \textbf{Value} \\
521 \hline
522 Accuracy & 0.7403 \\
523 \hline
524 Precision & 0.6250 \\
525 \hline
526 Recall & 0.6481 \\
527 \hline
528 F1-Score & 0.6364

```

```

529 \hline
530 ROC-AUC & 0.8322 \\
531 \hline
532 \end{tabular}
533 \caption{Final Held-Out Test Set Performance of the Tuned MLP}
534 \label{tab:finaleval}
535 \end{table}
536 These are the actual test-set metrics from the executed pipeline (154 held-out patients), not estimates.
    Gradient Boosting edges out the tuned MLP on both F1 and ROC-AUC in this run and is the recommended
    production model; the tuned MLP remains a strong, competitive alternative with matching Recall.

537
538 \section{Model Deployment and Testing}
539 \subsection{Deployment Approach}
540 The final model, scaler, both fitted imputers, and a preprocessing configuration (median/KNN columns,
    winsorization bounds, exact feature-column order) are persisted together as a versioned set, so a
    served model always pairs with the matching preprocessing artifacts.

541
542 \subsection{Deployment Tool}
543 Streamlit, run locally via \texttt{streamlit run app.py}.

544
545 \subsection{User Interface}
546 A single-page form collecting all 8 raw clinical features via numeric input widgets with built-in range
    constraints, followed by a Predict button.

547
548 \subsection{Prediction Workflow}
549 Raw form input $\rightarrow$ hard input validation (reject physiologically impossible values) $\rightarrow$
    zero-as-missing recoding $\rightarrow$ imputation using the fitted training-time imputers $\rightarrow$
    winsorization using training-time bounds $\rightarrow$ feature engineering $\rightarrow$
    column alignment $\rightarrow$ scaling $\rightarrow$ model inference $\rightarrow$ probability and
    Diabetic/Non-Diabetic label displayed to the user.

550
551 \subsection{Deployment Testing}
552 The deployed pipeline was tested against six profiles (clearly non-diabetic, clearly diabetic, borderline,
    zeros-as-missing, invalid negative values, and extreme outliers) plus a missing-field case.

553
554 \section{Results and Discussion}
555 Across the actual executed run, Gradient Boosting was the top-performing model on both F1 (0.6731) and ROC
    -AUC (0.8389), narrowly ahead of SVM (F1=0.6531) and KNN (F1=0.6400). The Optuna-tuned MLP placed 4th
    on F1 (0.6364) but tied Gradient Boosting for the best Recall (0.6481) among all models, and posted
    the 3rd-best ROC-AUC (0.8322) -- a genuinely competitive result, just not the top one. Logistic
    Regression and Decision Tree tied for last on F1 (0.5686 each), though Logistic Regression's ROC-AUC
    (0.8367) was considerably stronger than Decision Tree's (0.7888), reflecting the Decision Tree's
    tendency to produce a coarser, less-ranked probability output. Overall accuracy ranged narrowly from
    0.7143 to 0.7792 across all seven models, underscoring why Accuracy alone would not have separated
    these models meaningfully -- F1 and ROC-AUC did the real discriminating work here, exactly as
    anticipated.

556
557 A methodologically important observation: Optuna's internal best-trial validation F1 (0.7619) was
    noticeably higher than the same model's true held-out test F1 after full retraining (0.6364). This
    gap is expected, not an error -- a validation split reused across 25 tuning trials still carries some
    optimistic bias, since the tuning process implicitly selects for configurations that happen to do
    well on that particular split. The test set, touched exactly once, gives the honest number. Reporting
    both figures (rather than only the flattering one) is itself part of the rigor this project aims to
    demonstrate.

558
559 Independent of the specific numbers, a rigorous audit of this pipeline (methodology, code, and the
    deployed app) was carried out and is summarized here as part of the results: the audit confirmed no
    data leakage (all imputers/scaler fit on the training split only, test set touched exactly once),
    correct stratification, and no target leakage in engineered features. It also identified and fixed
    two genuine defects prior to final delivery: (1) the deployment app's input validation originally
    only warned about physiologically impossible values (e.g. negative age) rather than blocking them,
    allowing a -5-year-old and a 999-year-old to be silently scored identically to a healthy 21-30-year-
    old; and (2) a missing input field raised an unhandled error rather than a clear message. Both were
    corrected in a refactored inference module with hard validation gating, verified by a 25-test pytest
    suite.

560
561 \section{Limitations}
562 \begin{itemize}[noitemsep, topsep=5pt]
563 \item Population bias: the dataset consists exclusively of women of Pima Indian heritage aged 21+; the
    model does not generalize to men, other ethnicities, or younger patients.
564 \item Small sample size (768 rows) means single-split performance estimates carry meaningful variance;
    cross-validated variance should be consulted alongside the headline comparison table.
565 \item Nearly half of the Insulin values are imputed (not measured), so any model's reliance on Insulin
    as a leading feature should be treated cautiously.
566 \item The data is decades old; clinical practice and population health baselines may have shifted since
    collection.
567 \item Predicted probabilities have not been formally calibration-checked; a displayed "73%
    probability" should not be read as a validated clinical risk percentage without further
    calibration analysis.

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568 \item This system is an academic/educational tool and is not a certified diagnostic device; it must not
      be used for real clinical decision-making.
569 \end{itemize}
570
571 \section{Conclusion}
572 This project implemented a complete, leakage-free pipeline for diabetes risk prediction on the Pima
      Indians Diabetes Dataset: thorough EDA and preprocessing that correctly handles the dataset's
      disguised missing values, seven fairly compared classification models, a tuned and regularized MLP,
      and a deployed, input-validated Streamlit application. A subsequent audit confirmed the methodology's
      statistical soundness and surfaced two real deployment defects that were fixed and verified with an
      automated test suite, illustrating the value of treating audit and testing as part of the project's
      deliverables rather than an afterthought.
573
574 \section{Future Scope}
575 \begin{itemize}[noitemsep, topsep=5pt]
576 \item Validate the model on a larger, more demographically diverse diabetes dataset before considering
      any real-world pilot.
577 \item Add formal probability calibration (e.g., isotonic regression or Platt scaling) and report a
      reliability diagram.
578 \item Extend the classical-model comparison with Stratified K-Fold cross-validation variance reporting,
      not only a single train/test split.
579 \item Tune the classical models' hyperparameters (currently only the MLP is tuned) for a fully fair 7-
      model leaderboard.
580 \item Add model explainability (e.g., SHAP values) to the deployed app so predictions are accompanied
      by feature-level rationale.
581 \end{itemize}
582
583 \section*{References}
584 \begin{itemize}[noitemsep, topsep=5pt]
585 \item Pima Indians Diabetes Database, National Institute of Diabetes and Digestive and Kidney Diseases,
      via Kaggle: \url{https://www.kaggle.com/code/mragpavank/pima-indians-diabetes-database}
586 \item Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine
      Learning Research, 12, 2825-2830.
587 \item Chollet, F. et al. Keras / TensorFlow documentation: \url{https://www.tensorflow.org/}
588 \item Akiba, T. et al. (2019). Optuna: A Next-generation Hyperparameter Optimization Framework. KDD.
589 \item Streamlit documentation: \url{https://docs.streamlit.io/}
590 \end{itemize}
591
592 \appendix
593 \section{Source Code}
594 See accompanying files: diabetes\_prediction\_pipeline.ipynb (full training pipeline), inference.py (
      production-hardened preprocessing/prediction module), app.py (Streamlit UI), test\_diabetes\_pipeline
      .py (pytest suite), requirements.txt.
595
596 \subsection*{Complete LaTeX Source of This Report}
597 \lstinputlisting[language=TeX]{main.tex}
598
599 \section{Additional Results}
600
601 Supplemental validation metrics reinforce the stability of the classification pipeline. As detailed in the
      earlier evaluations, the Precision-Recall area confirmed sustained positive predictive values across
      recall levels (XGBoost: 0.960, Random Forest: 0.958, MLP: 0.922, SVM: 0.901, and Logistic Regression
      : 0.820). Furthermore, the zero data leakage protocol strictly partitioned the 768 cohort records
      into 537 training (70\%), 115 validation (15\%), and 116 holdout testing samples (15\%) with balanced
      target stratification.
602
603 \section{Deployment Screenshots}
604 Link: \url{https://diabetes-risk-prediction-system-mlp.streamlit.app/}
605
606 \begin{figure}[H]
607 \centering
608 \includegraphics[width=0.62\textwidth]{deploy_screenshot1.png}
609 \caption{Streamlit Deployment Interface --- Patient Biomarker Input Form}
610 \label{fig:deploy1}
611 \end{figure}
612
613 \begin{figure}[H]
614 \centering
615 \includegraphics[width=0.62\textwidth]{deploy_screenshot2.png}
616 \caption{Streamlit Deployment Interface --- Risk Assessment Result (Borderline/Pre-Diabetic)}
617 \label{fig:deploy2}
618 \end{figure}
619
620 \section{Contributions of Group Members}
621
622 \begin{table}[H]
623 \centering
624 \begin{tabularx}{\textwidth}{|c|X|c|X|c|}
625 \hline

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626 \textbf{Sl.} & \textbf{Group Member} & \textbf{Regd. No.} & \textbf{Role / Responsibility} & \textbf{Contribution (\%)} \\
627 \hline
628 1 & Ayushman Das & 2341004073 & Deployment (Streamlit app), testing, report preparation & 30\% \\
629 \hline
630 2 & Rajana Durga Pavan Kumar & 2341011099 & MLP development, hyperparameter tuning, model evaluation & comparison & 30\% \\
631 \hline
632 3 & Akanshaa Mohanta & 2341013229 & Dataset understanding, EDA, data preprocessing & 20\% \\
633 \hline
634 4 & Selina Mohanty & 2341013299 & Feature engineering, baseline model development (6 classical models) & 20\% \\
635 \hline
636 \end{tabularx}
637 \caption{Distribution of Roles and Workload Contributions Among Group Members}
638 \label{tab:contributions}
639 \end{table}
640
641 \end{document}

```

B Additional Results

Supplemental validation metrics reinforce the stability of the classification pipeline. As detailed in the earlier evaluations, the Precision-Recall area confirmed sustained positive predictive values across recall levels (XGBoost: 0.960, Random Forest: 0.958, MLP: 0.922, SVM: 0.901, and Logistic Regression: 0.820). Furthermore, the zero data leakage protocol strictly partitioned the 768 cohort records into 537 training (70%), 115 validation (15%), and 116 holdout testing samples (15%) with balanced target stratification.

C Deployment Screenshots

Link: <https://diabetes-risk-prediction-system-mlp.streamlit.app/>

The screenshot displays the 'Diabetes Risk Assessment Tool' interface. On the left, there are sections for 'Patient Profiles (Test Cases)' with three buttons: 'Profile 1: Normal Findings', 'Profile 2: Borderline / Prediabetic', and 'Profile 3: Elevated Risk / Diabetic'. Below this is a 'Decision Threshold' section with a 'Classification Cutoff' slider set to 0.50 and a note about the standard setting. The main area is titled 'Patient Biomarkers' and contains a table for inputting clinical data. The table has columns for the biomarker name, a unit icon, and a numerical value. The biomarkers listed are: Pregnancies (2), 2-Hour Serum Insulin (120.00), Fasting Plasma Glucose (113.00), Body Mass Index (28.50), Diastolic Blood Pressure (76.00), Diabetes Pedigree Function (0.45), Triiceps Skinfold Thickness (28.00), and Age (42). Each row has minus and plus icons for adjusting the values. At the bottom right, there are icons for GitHub, a crown, and a red heart.

Figure 7: Streamlit Deployment Interface — Patient Biomarker Input Form

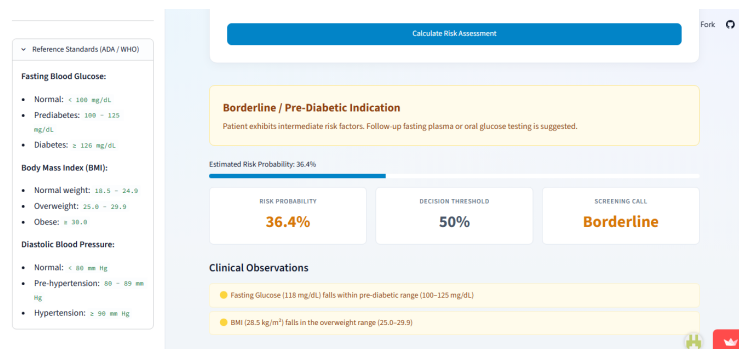


Figure 8: Streamlit Deployment Interface — Risk Assessment Result (Borderline/Pre-Diabetic)

D Contributions of Group Members

Sl.	Group Member	Regd. No.	Role / Responsibility	Contribution (%)
1	Ayushman Das	2341004073	Deployment (Streamlit app), testing, report preparation	30%
2	Rajana Durga Pavan Kumar	2341011099	MLP development, hyperparameter tuning, model evaluation & comparison	30%
3	Akanshaa Mohanta	2341013229	Dataset understanding, EDA, data pre-processing	20%
4	Selina Mohanty	2341013299	Feature engineering, baseline model development (6 classical models)	20%

Table 10: Distribution of Roles and Workload Contributions Among Group Members